

Respiratory Outbreak Simulator

1. Project Title

Respiratory Outbreak Simulator: Modelling the Impact of Public Health Interventions

2. Research Question

How do different public health interventions affect the simulated spread of a respiratory infectious disease in a synthetic population?

3. Aim

To develop a computational SIR model in Python to simulate the spread of a respiratory infectious disease and compare the effects of different public health intervention scenarios.

4. Study Population

A synthetic population of **5,000 individuals** will be used.

5. Model

SIR Model

- **S = Susceptible:** people who can become infected
- **I = Infectious:** people who are currently infectious
- **R = Recovered/Removed:** people who are no longer infectious under the model assumptions

6. Initial Population

- Susceptible = **4,990**
- Infectious = **10**
- Recovered/Removed = **0**
- Total = **5,000**

7. Baseline Scenario

The first simulation will represent an outbreak **without any intervention**.

This baseline will later be compared with intervention scenarios such as vaccination and transmission reduction.

8. Data

The project will use **synthetic data** created for modelling purposes and will not use identifiable patient data.

9. Model Parameters

Transmission parameter (β): 0.30 per day

This represents the illustrative rate at which infection spreads through the susceptible population under the model assumptions.

Recovery/removal parameter (γ): 0.20 per day

This represents the illustrative rate at which infectious individuals leave the infectious compartment.

Approximate average infectious period: 5 days

Calculated as:

$$1/\gamma = 1/0.20 = 5 \text{ days}$$

Initial reproduction number (R_0): 1.5

Calculated as: $R_0 = \beta/\gamma = 0.30/0.20 = 1.5$

10. SIR Mathematical Model

The population is divided into three compartments:

S = Susceptible

I = Infectious

R = Recovered/Removed

The model describes how people move between these compartments:

$$S \rightarrow I \rightarrow R$$

The three equations are:

$$\begin{aligned}\frac{dS}{dt} &= -\beta \frac{SI}{N} \\ \frac{dI}{dt} &= \beta \frac{SI}{N} - \gamma I \\ \frac{dR}{dt} &= \gamma I\end{aligned}$$

Plain-English explanation**Equation 1 - Susceptible**

$$\frac{dS}{dt} = -\beta \frac{SI}{N}$$

This calculates how many susceptible people are becoming infected.

Equation 2 - Infectious

$$\frac{dI}{dt} = \beta \frac{SI}{N} - \gamma I$$

This calculates how the number of infectious people changes.

It has two parts:

new infections – recoveries

Equation 3 - Recovered

$$\frac{dR}{dt} = \gamma I$$

This calculates how many infectious people move into the recovered/removed group.

Now let's actually use the numbers

This is the important part.

At the beginning of our simulation:

Variable	Value
S	4,990
I	10
R	0
N	5,000
β	0.30
γ	0.20

Now calculate new infections:

$$\beta \frac{SI}{N}$$

Put in our numbers:

$$0.30 \times \frac{4990 \times 10}{5000}$$

This gives approximately:

$$2.994$$

So the model says approximately 3 new infections per day at this particular starting point.

Now calculate recoveries:

$$\gamma I$$

$$0.20 \times 10 = 2$$

So approximately 2 infectious people recover/leave the infectious compartment per day at this starting point.

Therefore:

Susceptible

$$4,990 - 2.994 = 4,987.006$$

Infectious

$$10 + 2.994 - 2 = 10.994$$

Recovered

$$0 + 2 = 2$$

So after approximately one day, our simplified calculation gives:

State	Starting	After ~1 day
Susceptible	4,990	4,987.006
Infectious	10	10.994
Recovered	0	2
Total	5,000	5,000

This is what the computer will eventually do

It will repeat this process:

Calculate new infections → calculate recoveries → update S/I/R → repeat
for day 2, day 3, day 4, and so on.

11. Simulation Period

The SIR model will simulate the respiratory disease outbreak over a period of 180 days.

The model will calculate the number of susceptible, infectious and recovered/removed individuals at successive time points throughout the simulation.

Then Point 12 - Simulation Scenarios

We need something to compare.

We'll eventually run the model four times:

Scenario 1 - Baseline

No intervention.

5,000 people



Disease spreads



Observe outbreak

Scenario 2 - Vaccination

We will introduce an illustrative vaccination scenario.

Scenario 3 - Transmission reduction

We will model an intervention that reduces transmission, representing measures such as masking or reduced contact.

Scenario 4 - Combined intervention

Vaccination + transmission reduction.

12. Simulation Scenarios

Four scenarios will be simulated:

- 1. Baseline: No public-health intervention.**
- 2. Vaccination: An illustrative vaccination intervention.**
- 3. Transmission reduction: An intervention that reduces the effective transmission rate.**
- 4. Combined intervention: Vaccination combined with transmission reduction.**

The baseline scenario will provide the reference point against which intervention scenarios will be compared.

13. Model Outcomes

The model will be used to generate and compare the following outcomes:

- 1. Peak infectious population – the maximum number of infectious individuals at the same time.**
- 2. Day of epidemic peak – the day on which the infectious population reaches its maximum.**
- 3. Cumulative infections – the total number of individuals who have moved through the infection process during the simulation.**
- 4. Final susceptible population – the number of people remaining susceptible at the end of the simulation.**

5. Epidemic curves – graphical representation of changes in susceptible, infectious and recovered/removed populations over time.
6. Scenario comparison – comparison of outcomes between the baseline and intervention scenarios.

14. Baseline Simulation Results

The baseline SIR simulation was run for 180 days using the illustrative model parameters $\beta = 0.30$ per day and $\gamma = 0.20$ per day.

The simulated epidemic reached a peak infectious population of 330.729 individuals on Day 52.

By Day 180, the model contained approximately:

- 2,042.84 susceptible individuals
- 0.036 infectious individuals
- 2,957.12 recovered/removed individuals

These results represent the output of the mathematical model under the specified assumptions and should not be interpreted as a forecast of a real-world outbreak.

15. Vaccination Scenario

A vaccination scenario will be simulated using an illustrative baseline vaccination coverage of 40% and an assumed vaccine effectiveness of 80%.

The effective protected population will therefore be calculated as:

$$5000 \times 0.40 \times 0.80 = 1600$$

The vaccination scenario will then be compared with the baseline scenario using peak infectious population, timing of the epidemic peak and cumulative simulated infections.

16. Baseline vs Vaccination Results

The baseline simulation produced a peak infectious population of **330.729 individuals on Day 52**.

The vaccination scenario produced a peak infectious population of **10.494 individuals on Day 28**.

The simulated peak infectious population was therefore approximately **96.8% lower** under the vaccination scenario compared with the baseline.

These findings represent a comparison under the specified model assumptions and should not be interpreted as real-world estimates of vaccine effectiveness.

17. Transmission Reduction Scenario

A transmission-reduction intervention will be introduced from Day 30 of the simulation.

The intervention will reduce the effective transmission parameter by 50%, giving:

$$\beta_{\text{intervention}} = 0.30 \times 0.50 = 0.15$$

The scenario will be compared with the baseline and vaccination scenarios using the same model outcomes.

This intervention effect is an illustrative modelling assumption and is not intended to represent the effectiveness of a specific real-world intervention.